ARELLANO UNIVERSITY MALABON · ELISA ESGUERRA CAMPUS

LabGuard Smart Locker System

Automated Raspberry Pi 3 Laboratory Equipment Management with Frictionless QR-to-Mobile Authentication

PROTOTYPE SYSTEM v1.1

Python FastAPI · React 18 · SQLite · GPIO

Client Showcase & Software Specification

01. PROJECT OVERVIEW & CLIENT VALUE

Replacing Manual Logbooks & Keys with Smart Accountabili

LabGuard automates laboratory equipment borrowing and returning at Arellano University Malabon. Powered by a Raspberry Pi 3 controlling 10 physical solenoid locker relays, students and faculty authenticate on their personal phones via school email (@arellano.edu.ph) while admins monitor real-time circulation and overdue alerts.

0 PINs	< 1 sec	10 Units
QR + Email Auth	WebSocket Push	GPIO Relays

02. END-TO-END QR-TO-LOCKER AUTHENTICATION & HARDWARE FLOW

100% Offline Local Lab Wi-Fi Operation

STEP 1: SCAN KIOSK QR

Dynamic 5-Min Session QR

Touchscreen LCD generates a single-use QR token (/mobile/login/{token}) and opens a WebSocket listener.

STEP 2: VERIFY AU EMAIL

School Email Validation

Borrower enters @arellano.edu.ph email on phone. FastAPI validates domain and pushes auth event to LCD.

STEP 3: LOCKER UNLOCK

10s GPIO Relay Actuation

Selecting item energizes Raspberry Pi BCM GPIO relay for 10s, logs due date, and updates Admin tracker.

GPIO Relay Pins (BCM): 17, 27, 22, 5, 6, 13, 19, 26, 12, 16 | Status LEDs: GPIO 24 (Ready), GPIO 25 (Fault) |

03. ACTUAL WEBAPP INTERFACE SNAPSHOTS — USER TOUCHSCREEN KIOSK, MOBILE EMAIL LOGIN & ADMIN DASHBOARD

SNAPSHOT A: USER TOUCHSCREEN KIOSK (/kiosk)

7" RPi Touch LCD

LabGuard Kiosk · Juan Dela Cruz (STUDENT)

juan.delacruz@arellano.edu.ph · Idle: 58s

Logout

Borrow Equipment | Return Equipment (2)

All | Glassware | Tools | Instruments | Optics | Safety

LKR #1	Pyrex Borosilicate Volumetric Flask Se… — Glassware · Tap to Borrow!	Avail (9)
LKR #2	Rigol DS1054Z 4-Channel 50MHz Digital … — Instruments · Tap to Borrow!	Avail (4)
LKR #3	Olympus CX23 Binocular Compound Micros… — Optics · Tap to Borrow!	Avail (6)
LKR #4	Fluke 87V Industrial True-RMS Digital … — Tools · Tap to Borrow!	Avail (8)

SNAPSHOT B: MOBILE AUTH

LabGuard Mobile Login

Arellano University Malabon

Enter AU School Email:

juan.delacruz@arellano.edu.ph

Confirm Login!

' Logged In Confirmed!

You may now use the kiosk.

WebSocket pushed to LCD.

Whitelist: @arellano.edu.ph

Single-Use 5m Token

SNAPSHOT C: ADMIN WEB DASHBOARD (/admin)

8 Admin Modules

http://192.168.1.50:8000/admin · Prof. Maria Cristina Santos

Active Borrows	Overdue Items	AU Borrowers	Lockers Ready
5	2	8	10 / 10

Borrower & Email	Equipment & Locker	Status
Juan Dela Cruz juan.delacruz@arellano.edu.ph	1x Rigol DS1054Z 4-Channel 50… Locker #2 · Due 2026-11-10	BORROWED
Juan Dela Cruz juan.delacruz@arellano.edu.ph	2x Fluke 87V Industrial True-… Locker #4 · Due 2026-11-08	OVERDUE
Aiko Michelle Reyes aiko.reyes@arellano.edu.ph	2x Olympus CX23 Binocular Com… Locker #3 · Due 2026-11-11	BORROWED
Marco Antonio Bautista marco.bautista@arellano.edu.ph	3x Pyrex Borosilicate Volumet… Locker #1 · Due 2026-11-05	OVERDUE
Engr. Roberto M. Diaz roberto.diaz@arellano.edu.ph	3x Hanna HI98129 Waterproof p… Locker #8 · Due 2026-11-14	BORROWED

CONFIRMED TECH STACK: Python 3.11 + FastAPI · SQLAlchemy + SQLite · React 18 + Tailwind CSS · WebSockets · APScheduler · Raspberry Pi 3 gpiozero

Arellano University Malabon — LabGuard Smart Locker Prototype v1.1 | User Kiosk (/kiosk) · Mobile Login (/mobile/login/{token}) · Admin Portal (/admin)

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LabGuard Smart Locker

Software & Hardware Prototype Specification Pubmat

RASPBERRY PI 3 + PYTHON FASTAPI

1. USER-FACING TOUCHSCREEN KIOSK WORKFLOW (9 SCREENS)

1. Idle / Welcome Screen

AU Malabon branding, live digital clock, full-width "TAP TO BEGIN" CTA card.

2. QR Login Screen

280x280px single-use session QR code, 5-min expiry timer, WebSocket listener.

3. Authenticating Screen

Waiting for mobile email verification with 30-second timeout protection.

4. Equipment Catalog

Large Borrow/Return touch tabs, category filters, On-Screen QWERTY Tap Keyboard.

5. Item Detail & Borrow

Large +/- quantity buttons (>=48px) and Due Date presets (1 Day, 3 Days, 1 Week).

6. Opening Locker Screen

Triggers Raspberry Pi BCM GPIO relay for 10s with fault timeout detection.

7. Borrow Confirmed

Displays item summary, Locker #, security reminder, and 15s auto-return to Idle.

8. Return Equipment Tab

Lists user active borrows with overdue badges and multi-item tap checkboxes.

9. Return Confirmed

Unlocks locker for return and displays AU Red late-return banner if overdue.

2. ADMIN WEB DASHBOARD (/admin) — 8 MANAGEMENT MODULES

1. Dashboard Overview

5 KPI cards (Borrowed, Overdue, Returned Today, Users, Equipment) + Live Feed.

2. Borrowed Tracker

Sortable Blue/Red/Green color-coded rows, 25/50/100 pagination & manual return override.

3. Return History

Full log of completed returns with loan duration and On-Time (Yes/No) compliance.

4. Overdue Items Panel

Urgency color-coding: Yellow (1–2 days past due) and Red (3+ days past due).

5. Equipment Inventory

Add/edit items, assign Lockers #1–#10, +/- stock adjustments with audit log.

6. User Management

Manage Student/Faculty @arellano.edu.ph accounts + one-click CSV bulk import.

7. Reports & Export

Export PDF & CSV reports (Borrow Frequency, Active Borrowers, Overdue, Compliance).

8. System Settings

Configure whitelisted email domains, QR expiry, due date presets & admin users.

3. SMART LOCKER BANK & RASPBERRY PI 3 BCM GPIO MAPPING

Locker	BCM GPIO	Assigned Laboratory Equipment	Category	Stock
#1	GPIO 17	Pyrex Borosilicate Volumetric Flask Set (250mL & 500	Glassware	9/12
#2	GPIO 27	Rigol DS1054Z 4-Channel 50MHz Digital Oscilloscope	Instruments	4/6
#3	GPIO 22	Olympus CX23 Binocular Compound Microscope	Optics	6/8
#4	GPIO 5	Fluke 87V Industrial True-RMS Digital Multimeter	Tools	8/10
#5	GPIO 6	Eppendorf Research Plus Adjustable Micropipette (100	Instruments	10/10
#6	GPIO 13	Corning Graduated Cylinder & Burette Titration Kit (	Glassware	7/8
#7	GPIO 19	Hakko FX-888D Digital Soldering & Rework Station	Tools	5/6
#8	GPIO 26	Hanna HI98129 Waterproof pH / Conductivity / TDS Tes	Instruments	2/5
#9	GPIO 12	3M Chemical Splash Goggles & Nitrile Hazmat Apron Ki	Safety	14/15
#10	GPIO 16	Vernier Digital Caliper & Micrometer Precision Set	Tools	0/6